THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Final Exam, 2022

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Reading time: 30 minutes — Writing time: 3 hours — Upload time: 30 minutes

This exam consists of 19 pages (including this page) with 11 questions and 106 total marks

Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier and blank loose-leaf paper.
- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted. No headphones or earphones are permitted.

Instructions to Students

- Wave your hand right in front of your webcam if you wish to communicate with the supervisor at any time (before, during or after the exam).
- You must not be out of webcam view at any time without supervisor permission.
- You must not write your answers on an iPad or other electronic device.
- Off-line PDF readers (i) must have the screen visible in Zoom; (ii) must only be used to read exam questions (do not access other software or files); (iii) must be set in flight mode or have both internet and Bluetooth disabled as soon as the exam paper is downloaded.

Writing

- If you are writing answers on the exam or masked exam and need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- **You must not leave Zoom supervision to scan your exam.** Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned exam as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary. Do not leave Zoom supervision until you have confirmed orally with the supervisor that you have received the Gradescope confirmation email.
- **You must not submit or resubmit after having left Zoom supervision.**

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Question 1 (21 marks)

Short questions: In addition to your answer, you must show your working.

(a) Solve the following initial value problem

$$F(x) = \int \left(\sqrt[3]{x} + \cos(\pi x)\right)\, dx \quad \text{with } F(0) = 1$$

(b) You have 5 different pictures. How many different arrangements can be made by hanging them in a row, if at least three pictures are used for each arrangement?

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(c) Find the equation of the tangent line to $f(x) = e^{x^2+x}$ about $x = 0$.

(d) Let $f(x) = 3x^2 \log(6x)$. Find $f'(x)$.

(e) Evaluate the following:

$$\int \frac{x^5 + 2x^3 + x}{x^4} \, dx$$

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(f) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x \to 2} \frac{x^3 - 7x + 6}{x - 2}$$

(g) Consider the quadratic function $f(x) = x^2 + 5x + 2$. Write $f(x)$ in turning-point form (i.e. $f(x) = (x - h)^2 + k$) and hence determine the coordinates of its turning point.

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Extended problems: Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justification of your solutions.

Question 2 (13 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$, with $f(x) = 2\sin(2x) + 1$.

(a) State the domain and range of the function. You should justify your answer for the range - you may do so by indicating what graph transformations you need to transform $\sin(x)$ into the given function $f(x)$.

(b) Is $f(x)$ injective? Justify your answer.

(c) Find the y-intercept of $f(x)$.

(d) Find all the x-intercepts of the function $f(x)$.

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Question 3 (11 marks)

Consider the functions $f(x) = \sqrt{x+1}$ and $g(x) = x^2 + 2x - 1$.

(a) What is the maximal domain of $f(x)$? And what is the maximal domain of $g(x)$?

(b) Sketch $f(x)$, labelling any important point(s).

(c) Find the inverse of $f(x)$ and state its domain and range.

(d) Find an expression for $f(g(x))$.

(e) What is the maximal domain of $f(g(x))$?

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Question 4 (14 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \dfrac{2-x}{5+x^2}$.

(a) Find all x and y intercepts of the function.

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative.

(c) Evaluate and simplify $f'(x)$.

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(d) Find the location of the critical points of $f(x)$.

(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing.

(f) Find and classify the location of any local extrema of the function.

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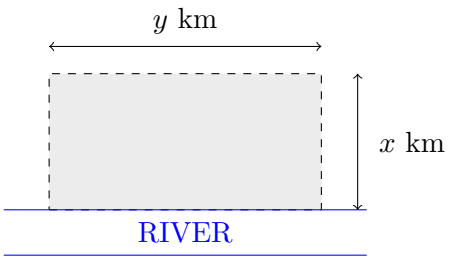
Question 5 (9 marks)

A farmer wants to use fencing wire to fence a rectangular piece of land, using a river as one of the sides (so no fencing is needed on that side), as shown in the picture below:



(a) What is the total length of fencing wire he will need for the job? (Give your answer in terms of x and y)

(b) The farmer wants to use a total of 4km of fencing wire for the job. Use this information to write an expression for y in terms of x.

(c) What is the area of the enclosed land in the situation described above? Leave your answer in terms of x only. (*You can assume that the farmer is using the 4km of fencing wire as in part (b)*)

(d) What are the dimensions x and y that give the maximum enclosed area? What is the maximum enclosed area?

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Question 6 (5 marks)

A melting snowball (which can be assumed to always be spherical in shape) is decreasing in volume at a constant rate of 8 cm^3/min. Find the rate at which the radius is changing when the snowball has a radius of 4cm. (*Hint: The volume of a sphere of radius r is* $V = \frac{4}{3}\pi r^3$)

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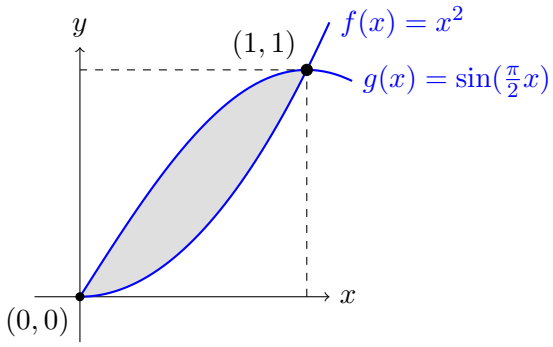
Question 7 (7 marks)

Consider the functions $f(x) = x^2$ and $g(x) = \sin\left(\frac{\pi}{2}x\right)$, with graphs as shown below:



(a) Show that the two functions intersect at $x = 0$ and at $x = 1$.

(b) Find the value of the shaded area.

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Question 8 (8 marks)

For this question, a decimal approximation is not required. You may leave your final answer in simplified fractional form.

The conditional probability formula says that, for two events $A, B \in \Omega$ with $P(A) \neq 0$,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

For any MAST10012 student, the probability that they forget the formula during the exam is $p = \frac{2}{3}$. We randomly choose 4 MAST10012 students and define X to be the number of those students that forget the formula during the exam.

(a) Calculate the probability that exactly 2 of those 4 students forget the conditional probability formula during the exam.

(b) What is the probability that at least one of the students forgets the formula during the exam?

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(c) What is the probability that exactly two students forget the formula during the exam, given that at least one student forgets it?

Question 9 (5 marks)

In how many ways can the letters of the word MATHS be arranged

(a) if all the letters are used?

(b) if we form 3-letter words?

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Question 10 (5 marks)

The probability density function for a random variable X is

$$f(x) = \begin{cases} 1 & a < x < 2 \\ 3 - x & 2 \leq x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that we must have $a = \frac{3}{2}$.

(b) Find $P(X < 2)$.

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Question 11 (8 marks)

The amount of water Mr. Green uses to water his garden is normally distributed with mean 100 litres and standard deviation 10 litres.

(a) On 20% of occasions, it takes him less than k litres of water to water the garden. What is the value of k?

(b) He wants to be environmentally friendly and aims to use between 90 and 95 litres to water his garden. What percentage of times is he hitting that goal?

End of Exam — Total Available Marks = 106

Turn the page for appended material

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Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

- **Functions:**
  - Lines

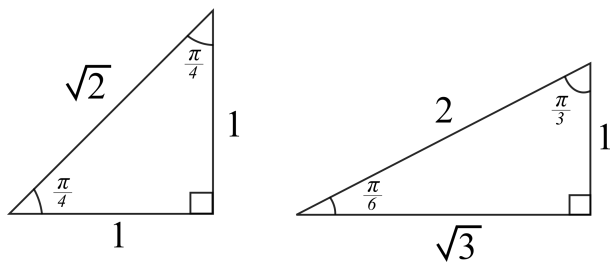
$$y = mx + c \qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**



- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$

$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$

$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$

$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$

$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$

$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$

$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$

$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$

$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$

$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$

$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

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- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a} [f(x) + g(x)] = K + L \qquad \lim_{x\to a} [f(x) - g(x)] = K - L$$

$$\lim_{x\to a} [f(x) \times g(x)] = K \times L \qquad \lim_{x\to a} \left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$

$$\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a} [f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$

$$\lim_{x\to a} cf(x) = cK$$

$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$

$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{a\to x} \frac{f(a) - f(x)}{a - x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

- **Binomial:** For $X \sim B(n,p)$:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$\mathbb{E}(X) = np$$

$$\mathrm{Var}(X) = np(1-p)$$

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The standard normal distribution table for $P(X < x)$

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000